

biomarkr

Monitor, Detect, Prevent

PROBLEM

Inflammation drives every major disease.

1.5B

people living with
inflammatory disease

3 in 5

deaths worldwide are
inflammation-driven

\$2.1T

annual global
healthcare cost

The WHO ranks inflammatory disease as the single greatest threat to human health today.

Cancer

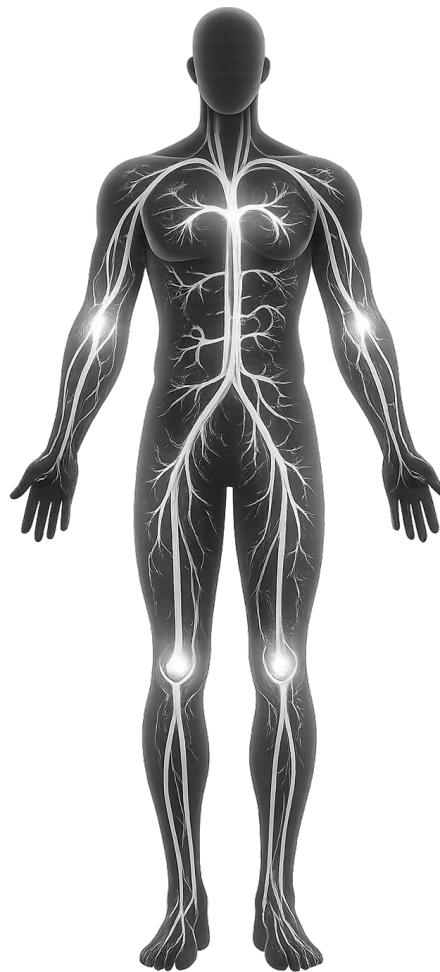
Cardiovascular disease

Autoimmune disease

Neurodegeneration

Metabolic syndrome

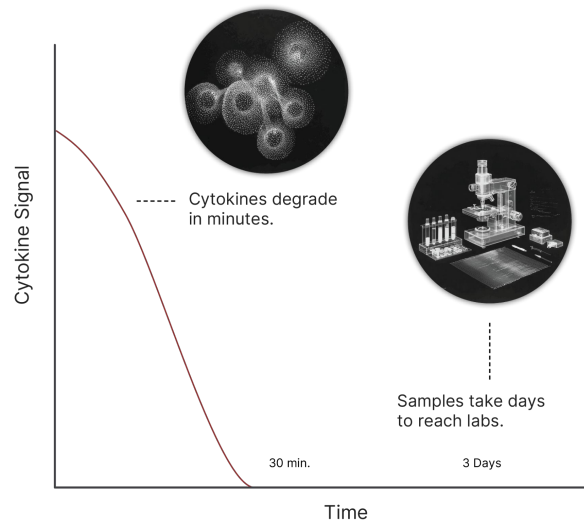
Aging



PROBLEM

Inflammation is uniquely hard to test.

The cytokines that signal inflammation are the hardest analytes in blood to measure. Tests are expensive, specialized, and almost never used in routine care.



01

Rapid degradation

Cytokines degrade within minutes of a draw — a delicate signal lost en route to a central lab.

02

Low concentrations

Markers circulate at vanishingly low levels, demanding sensitivity far beyond a standard blood panel.

03

Complex interpretation

A single value is meaningless in isolation. Meaning comes from a personal baseline and machine-learning analysis over time.

TECHNOLOGY

Our chip reinvents blood testing.

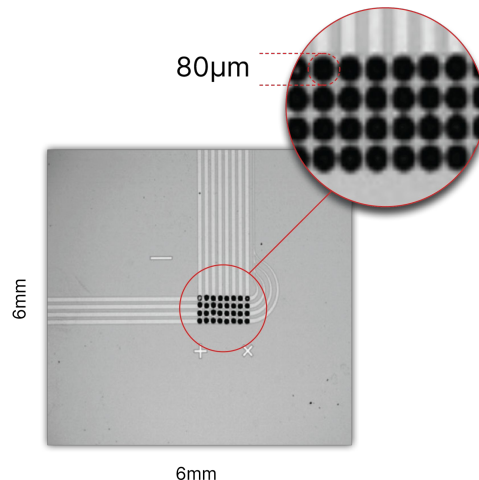
Solid-state silicon photonic chip: combining optics, fluidics, and signal detection on a single die.

Ultra-high sensitivity: whole blood, up to 32 multiplex markers, at low-cost manufacturing scale.

Lab-grade accuracy: proven on clinical samples, 55+ markers tested successfully.

22 patents granted: Development led by our CSO (Technion PhD), backed by BARDA and Roche.

Actual chip · 6mm × 6mm. Sensing wells appear as black dots, 80μm across.



SOLUTION

Test anywhere. Results in minutes.

A reader and a single-use cartridge turn a simple finger-prick into a lab-grade inflammation panel in minutes.



BENEFIT 01

Affordable

Price falls from \$550 to \$25, making frequent testing viable.

BENEFIT 02

Simple

Whole-blood testing, easy to use at home or point-of-care.

BENEFIT 03

Fast

Results in 10 minutes — before inflammation markers degrade.

BENEFIT 04

Repeatable

Frequent testing builds the longitudinal data AI analysis needs.

Our first panel tests six inflammation markers.

Signals largely invisible to central labs, measured together, on a baseline, over time.

IL-6

Interleukin-6

Early alarm signal driving fever and immune activation.

IL-10

Interleukin-10

The immune brake, suppressing excessive inflammation.

IL-1 β

Interleukin-1 Beta

Amplifies inflammation and recruits cells to injury.

IFN- γ

Interferon Gamma

Activates macrophages; coordinates antiviral immunity.

TNF- α

Tumor Necrosis Factor

Triggers cell death and drives systemic inflammation.

CRP

C-reactive Protein

Acute-phase protein, the clinical inflammation surrogate.

Keeping RA patients ahead of the flare.

STEP 1 · BASELINE

● A personal baseline

Maria, 44, tests her inflammation twice a week on a biologic.

STEP 2 · SIGNAL

● An early signal

IL-6 and TNF- α climb above baseline.
Maria still feels normal.

STEP 3 · ACTION

● A timely adjustment

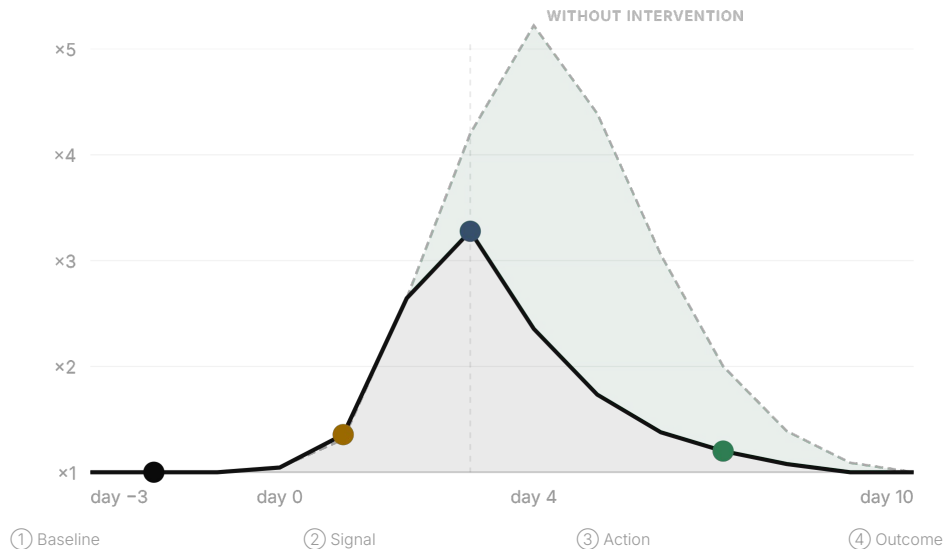
Her rheumatologist adjusts the regimen before symptoms appear.

STEP 4 · OUTCOME

● A blunted flare

Less joint damage, no urgent visit, no pain.

MARIA'S IL-6, MAPPED TO EACH STEP



Our moat compounds with every test.

MOAT 01

Technology

Price, speed, and sensitivity competitors can't match together.

MOAT 02

Intellectual property

22 patents + 2 pending across chip, optics, and assay.

MOAT 03 · THE CORE ASSET

Longitudinal data

GROWS WITH EVERY TEST →

The data layer is the only one that widens over time, becoming an impassable moat and our core asset.

VALIDATION

Milestones validate the technology.

22

Patents issued

Covering chip, optics, and assay.
With two provisional patents filed.

\$ 750k

R&D revenue + LOIs

A BARDA contract: "bringing the lab into the home" with letters of intent in place.

97.5%

Analytical accuracy

Measured against the Luminex gold standard on clinical samples.

One infrastructure, three growing markets.

The same testing infrastructure solves critical pain points across therapeutic development and chronic care before expands into wellness.

USER 01 · TODAY

Research (RUO)

Accelerates drug development and lowers the cost of clinical trials with rich, frequent cytokine data.

Live · revenue today

USER 02 · 510(k)

RA & IBD

Immune tracking supports autoimmune management and medication decisions in routine care.

Near-term · post-510(k)

USER 03 · DE NOVO

Wellness & longevity

Direct-to-consumer immune baselines for early prevention, at-home care, recovery, and healthy aging.

Expansion · DTC scale

Revenue scales with testing frequency.

PER-TEST ECONOMICS

\$25

Avg. price / test

\$3.60

Test COGS

\$21.40

Gross profit / test

\$400

Reader COGS (one-time)

12 tests / yr
per user



\$21M

24 tests / yr
per user



\$42M

52 tests / yr
per user



\$91M

Estimated annual revenue at 50,000 active users.

A \$61B market, entered through the clinic.

TAM
\$61.0B

Total addressable

SAM
\$17.3B

Serviceable addressable

SOM
\$163M

Obtainable by 2030 (5-yr)

CHANNEL	TAM	SAM	SOM · 5-YR
DTC — Wellness / Longevity	\$23B	\$7.0B	\$18M
DTC — Clinical / Subclinical	\$22B	\$6.0B	\$42M
B2B — Clinical	\$12B	\$3.2B	\$65M
RUO / Pharma	\$4.0B	\$1.1B	\$38M

Go-to-market follows regulatory milestones.



H2 2026

Pre-FDA · RUO sales



2027

510(k) approval



2029

De Novo approval



2030+

Panel expansion

TODAY · RUO

Accelerating trials

Selling to CROs and therapeutics developers to accelerate clinical trials. LOIs in place.

TAM \$4.0B

NEAR-TERM · CLINICAL

Specialty clinics

B2B sales to 12,000+ U.S. specialty clinics, focused on chronic management of RA & IBD.

TAM \$12B

EXPANSION · DTC

Wellness & longevity

B2B2C and direct-to-consumer sales for wellness and longevity users.

TAM \$45B

A de-risked, phased regulatory pathway.

Two completed FDA Q-Sub interactions de-risk our path to 510(k) clearance, followed by first-in-class De Novo classification.

CRP · 510(k)

Predicate-device route

COMPLETE 

Q-Sub interactions

2026 

Predicate identified

2027 

510(k) clearance

5-cytokine · De Novo

First-in-class classification

COMPLETE 

Analytical validation

2028 

Clinical validation

2029 

De Novo clearance

New panels expand revenue at lower regulatory risk.

- Inflammation monitoring is our market entry; the most differentiated panel, cleared first.
- Additional panels augment disease modeling, increase retention, and drive higher revenue per user.
- We've already proven the ability to test 55+ commonly used biomarkers.

Inflammation

Market entry · live

Cortisol

Stress & recovery

Cardiovascular

Risk & prevention

Hormones

Endocrine health

Therapeutic monitoring

Drug dosing

55+ markers

Validated & ready

From reactive snapshots to predictive data.

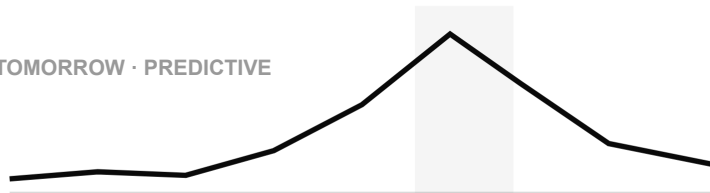
TODAY · REACTIVE



A single snapshot

One reading, taken when it already hurts.
Elevated inflammation says something is
wrong, but not what, and not when.

TOMORROW · PREDICTIVE



A longitudinal signal

A continuous baseline that identifies and
predicts disease before symptoms appear.
The foundation for prevention.

THE VISION

Inflammation data delivers real-time health insight.

IMMUNE BASELINE & AGE

A personal inflammage score

Frequent testing generates a personalized immune baseline and an inflammation-age score for each user.

EARLY DETECTION

An early-warning system

Change over time, compared to a user's baseline, surfaces disease and illness before symptoms.

LIFESTYLE & LONGEVITY

Actionable, every day

Inflammation reveals how recovery, diet, sleep, medication, and lifestyle shape overall health.



TEAM

Founded by expertise in AI, engineering, and medicine.



Dylan Brownstein

CHIEF EXECUTIVE OFFICER

2× founder (AI & media)

AI innovation @ Karcher Ventures

Experience @ Google & Reddit



Dr. Reuven Duer, PhD

CHIEF SCIENCE OFFICER

PhD from Technion Institute

Founder @ Proactive Diagnostics

Inventor of Q-SENS



Dr. Aren Giske, MD

CHIEF OPERATING OFFICER

2× Medical Director

Board Member @ Kadlec Medical

Occupational & Env. Medicine

THE ASK

\$12M

to complete regulatory submission and bring run-rate revenue past \$3.5M.

MILESTONE I

\$5.5M

Engineering finalized

Cost reduced, accuracy improved, manufacturing and regulatory ready.

Est. 6–12 months

MILESTONE II

\$1.5M

510(k) submitted

Predicate-device route for CRP clears the way for clinical go-to-market.

Est. 12–18 months

MILESTONE III

\$5M

Clinical partners

First 300 clinical partners supplement RUO revenue, bringing run-rate past \$3.5M.

Est. 20–24 months